

ELI — Architecture Ascii

ELI v2.1.81

August 2026

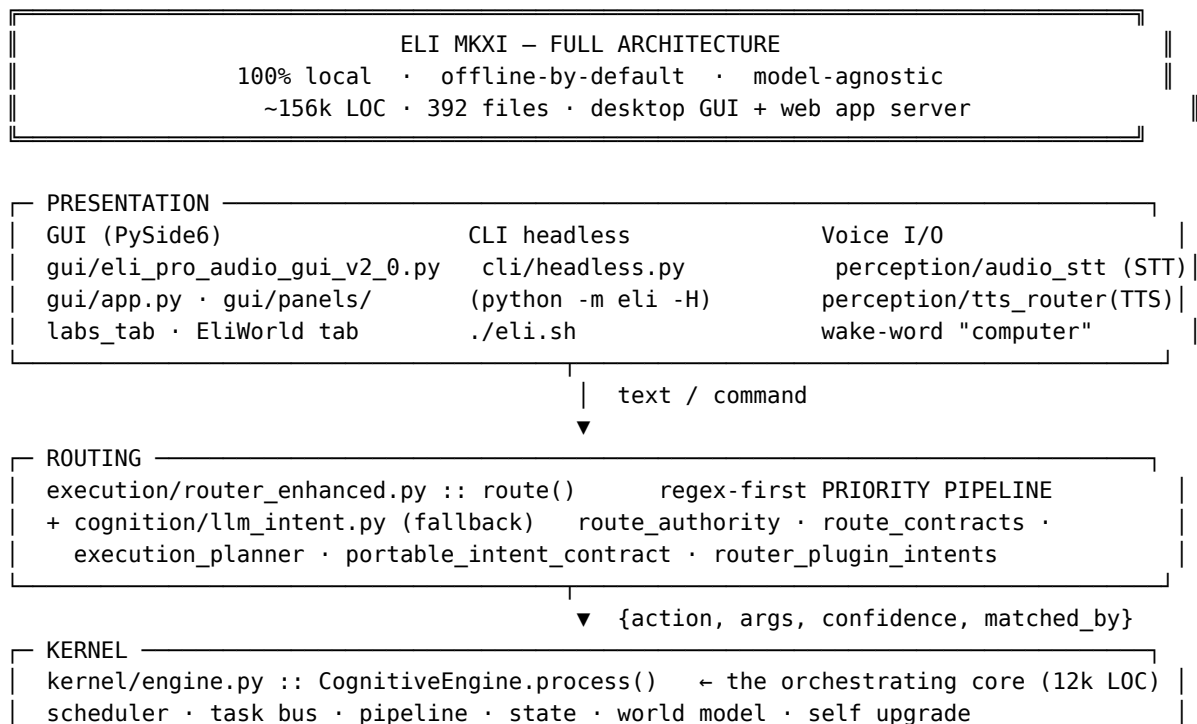
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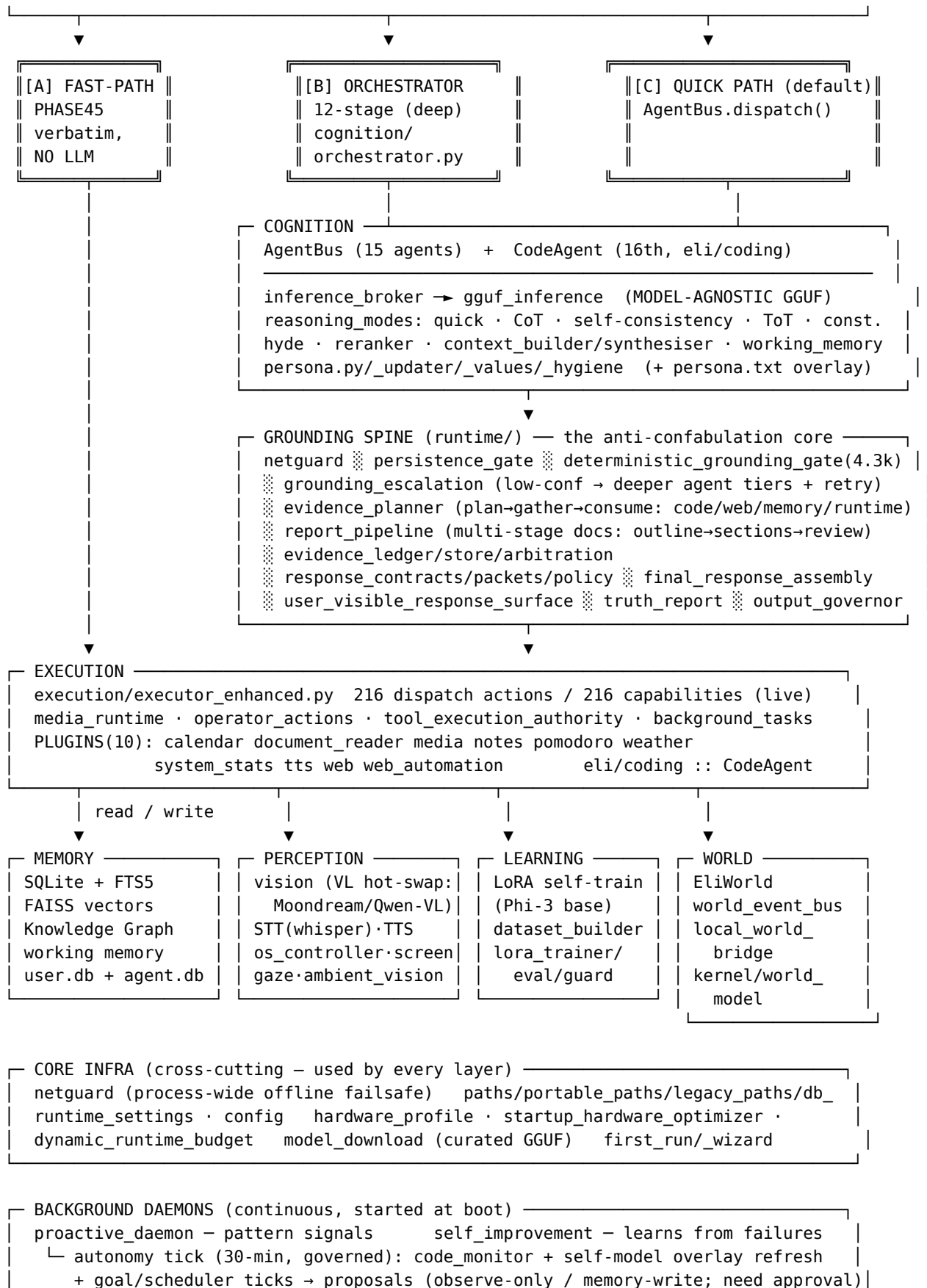
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Blueprint – ELI MKXI Full Architecture (ASCII)

The entire system in one drawing, plus the module tree and data layout. Grounded in the real source (see `architecture.md` for prose, `diagrams.md` for the pipeline/memory/gating close-ups). Every layer and box maps to a real path.

A. Full system — layered block diagram





habits_scheduler/habits – routines	scheduler · task_bus – jobs
background_tasks – async heavy work	reflection loop ambient_vision loop
scheduled_tasks (durable overnight/timed)	world_event_bus ← confidence events

```

OUTPUT ← output_governor / sanitiser ← inference ← (any path)
      |
      └─ TTS (Piper) / GUI render

```

B. The 15 agents

AGENT BUS (cognition/agent_bus.py :: AgentBus.dispatch → DispatchResult)
 selects a set → runs on a dependency DAG → aggregates grounding_confidence

BusMemoryAgent	recall: SQLite/FTS5/FAISS → context
KnowledgeGraphAgent	entities/relations from the KG
SystemAgent	runs executor actions (WEB_SEARCH, RUNTIME_AUDIT, ...)
OrchestratorAgent	plan / coordinate grounded synthesis
FileCodeAgent	searches the codebase for code-grounded answers
IntrospectionBus...	live runtime + gathers identity/awareness audits as evidence (persona summarises – never a data dump)
CapabilityAgent	capability manifest (what ELI can do)
ReflectionAgent	session reflection / insights
ProactiveAgent	proactive patterns / signals
HabitAgent	learned habits
SelfImprovementAgent	self-improvement state
FrontierAgent	frontier reasoning hooks
PluginAgent	dispatch to registered plugins
VoiceAgent	STT/TTS engine state
CriticAgent	second-tier verifier: checks retriever agents' output
+ CodeAgent (16th)	SEPARATE pipeline: plan→search→verify→repair
eli/coding/agent	(CODE_SOLVE / GENERATE_SCRIPT) – not a bus agent

C. Module tree (LOC · key files · role)

```

eli/ (~156k LOC, 392 files) · api/server.py (FastAPI web app + dashboard)
├── __main__.py ..... entry dispatch (GUI | --headless)
├── kernel/          13.1k — the core
│   ├── engine.py    12.0k  CognitiveEngine.process() – the spine *god-file
│   ├── scheduler.py      timed jobs
│   └── task_bus.py · pipeline.py · state.py · world_model.py · self_upgrade.py
├── execution/       20.5k — route → act
│   ├── executor_enhanced.py  15.0k  216 actions *god-file
│   ├── router_enhanced.py    6.1k  priority pipeline *god-file
│   ├── execution_planner.py · route_authority.py · route_contracts.py
│   ├── tool_execution_authority.py · operator_actions.py · operator_policy.py
│   └── media_runtime.py · router_plugin_intents.py · executor_plugin_handlers.py

```

- └─ portable_intent_contract.py · execution_intent_packets.py
- cognition/ 12.3k — think
 - └─ agent_bus.py 2.4k 15 agents + dispatch *
 - └─ orchestrator.py 12-stage deep retrieval
 - └─ gguf_inference.py 2.1k · inference_broker.py model-agnostic inference
 - └─ reasoning_modes.py · hyde.py · reranker.py · llm_intent.py
 - └─ persona.py/_updater/_values/_status/_hygiene (+ persona.txt, persona.auto.txt)
 - └─ context_builder.py · context_synthesiser.py · working_memory.py
 - └─ output_governor.py · response_governance.py · response_sanitizer.py
 - └─ grounded_status.py · introspection_agent.py · tone_analyzer.py · chat_model.py
- runtime/ 18.0k — grounding spine + daemons (68 files)
 - └─ deterministic_grounding_gate.py 4.3k * · grounding_escalation.py
 - └─ evidence_ledger/store/arbitration.py · memory_evidence.py
 - └─ persistence_gate.py · truth_report.py · control_contracts.py
 - └─ response_contracts/packets/policy.py · final_response_assembly/provider.py
 - └─ user_visible_response_surface.py · personal_memory_*.py · reflection.py
 - └─ background_tasks.py · self_improvement.py · code_monitor.py
 - └─ capability_sync.py · pending_proposal.py · runtime_policy.py
- memory/ 6.6k — remember (13 files)
 - └─ memory.py 4.5k Memory · SQLite + FTS5 *god-file
 - └─ vector_store.py FAISS index
- perception/ 5.4k — sense (18 files)
 - └─ vision.py · analyze_image/csv/pdfs/mesh.py · ambient_vision.py
 - └─ audio_stt.py · local_whisper_stt.py · voice_worker(_streaming).py · eli_listen.py
 - └─ tts_router.py · os_controller.py · screen_locator.py · gaze_engine.py
- planning/ 3.1k — proactivity (21 files)
 - └─ proactive_daemon.py · habits_scheduler.py · habits.py · jobqueue_cli.py
- coding/ 1.5k — CodeAgent (plan→search→verify→repair, 9 files)
- learning/ 3.1k — LoRA self-training (11 files)
 - └─ lora_trainer/eval/guard.py · dataset_builder/filters.py
 - └─ bootstrap_phi3_base.py · base_model_resolver.py · training_preflight.py
- plugins/ 1.9k — manager + 10 plugins
 - └─ calendar · document_reader · media · notes · pomodoro · weather · system_stats · tts · web · web_automation
- world/ 1.5k — EliWorld (world_event_bus, local_world_bridge)
- core/ 4.9k — infra
 - └─ netguard.py offline failsafe + allow_network()
 - └─ paths.py · portable_paths.py · legacy_paths.py · db_paths.py
 - └─ runtime_settings.py · config.py · grounding.py
 - └─ hardware_profile.py · startup_hardware_optimizer.py · dynamic_runtime_budget.py
 - └─ model_download.py · first_run.py · first_run_wizard.py
- gui/ 18.7k — PySide6 desktop
 - └─ eli_pro_audio_gui_v2_0.py 11.0k *god-file · app.py · labs_tab.py 5.7k
 - └─ panels/ (startup.py: model picker + FirstBootWizard, HardwareTuningDock)

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|
├─ tools/          8.1k — image_engine · news · document tools
├─ contracts/ 0.7k · cli/ 0.1k · system/ 0.3k · utils/ 0.9k

```

D. On-disk data layout

```

artifacts/
├─ db/
│   ├── user.sqlite3      memories(+_fts) · conversation_turns · conversations ·
│   │                     kg_entities(+_fts) · kg_relations · recall_log ·
│   │                     runtime_events · news_articles(+_fts) · news_reflections ·
│   │                     habits/habit_events/habit_rules · observations ·
│   │                     learning_replay · working_memory_pins · user_patterns ·
│   │                     session_summaries · corrections · failures
│   └─ agent.sqlite3      agent_dispatches · agent_metrics · improvements ·
│                           failures · code_patches · error_tracking · observations
├─ vectors/index.faiss    semantic index (embedder: nomic-embed-...Q4_K_M.gguf)
├─ conversations/ + archive/
├─ runtime/users/<uuid>/user_profile...    per-user profile
├─ runtime_snapshot.json    live model/runtime truth
├─ world/{snapshots,ledger,journal}/ · proactive/ · incidents/<date>.jsonl
├─ documents/ · scripts/ · analyze_image_*/
|
config/  settings.json (gitignored) · templates/settings.template.json ·
         settings.example.json · plugins_state.json
models/  <your>.gguf · embeddings/ · whisper/ · image/          voices/ *.onnx (Piper)

```

E. The flow in one line

```

INPUT → ROUTER → ENGINE ─┬─ FAST-PATH ───────────────────→ OUTPUT (verbatim)
                        │
                        │ ┌─ ORCHESTRATOR → retrieval ─┐
                        │ │ └─ QUICK → AGENT BUS ───┐
                        │ └────────────────────────┘
MEMORY (SQL/FTS5/FAISS/KG/WM) → context ←──────────┘
GATES: netguard · persistence · grounding · escalation · confidence · governor
INFERENCE (broker → gguf, model-agnostic) → OUTPUT → TTS/GUI
BACKGROUND: proactive · self-improve · habits · learning(LoRA) · world · scheduler

Companion docs: architecture.md (prose, every subsystem) · diagrams.md (pipeline / memory
/ gating close-ups) · tools/eval/README.md (the measurement layer). ""

```